



September 2nd, 2026

Continuum-Kinetic Models in Hypersonic Transitional Flows (University of Nottingham, UK)

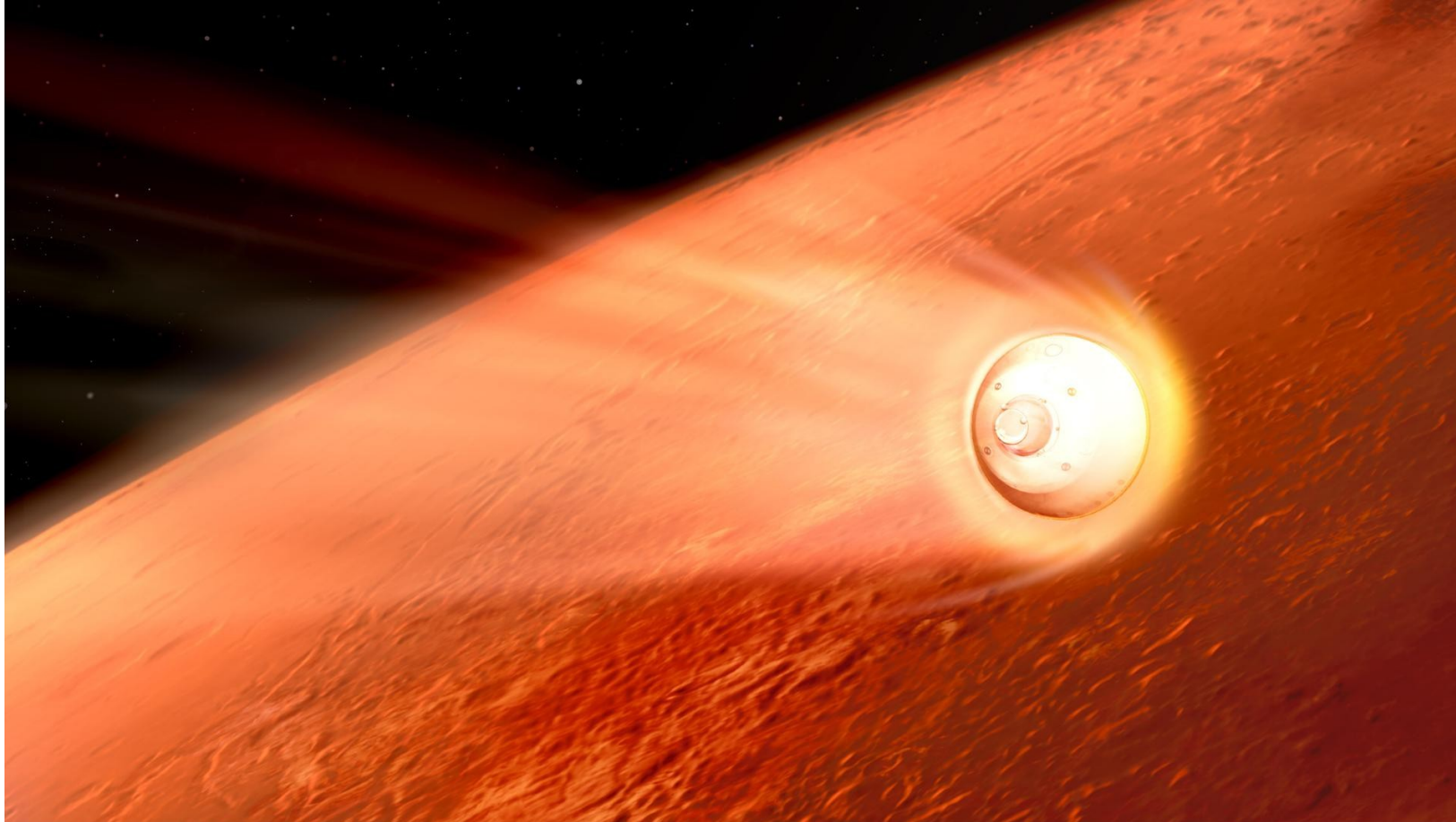
Modeling and simulation of hypersonic re-entry: challenges and perspectives

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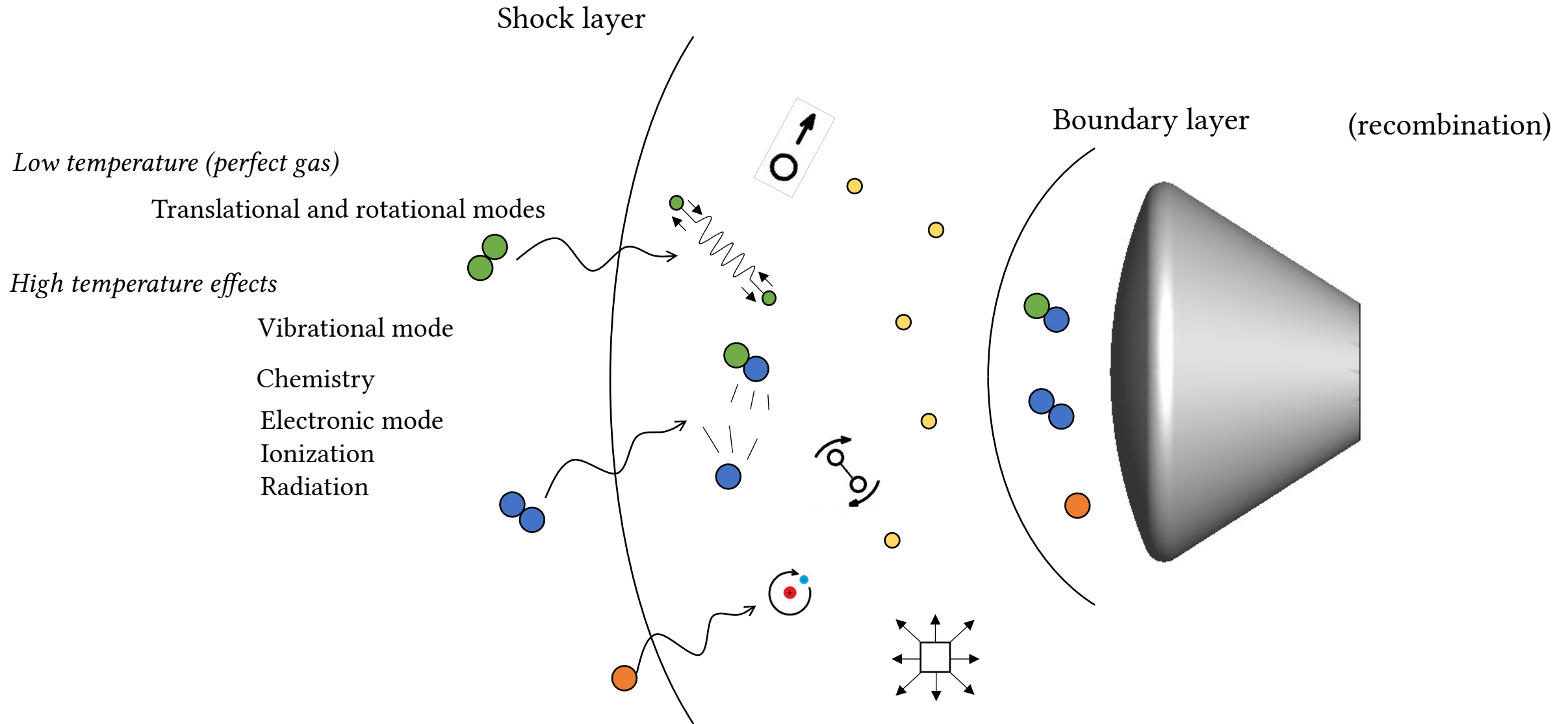
Introduction



Average household energy
consumption in Italy

It can power your house for
about 18 years!

Introduction



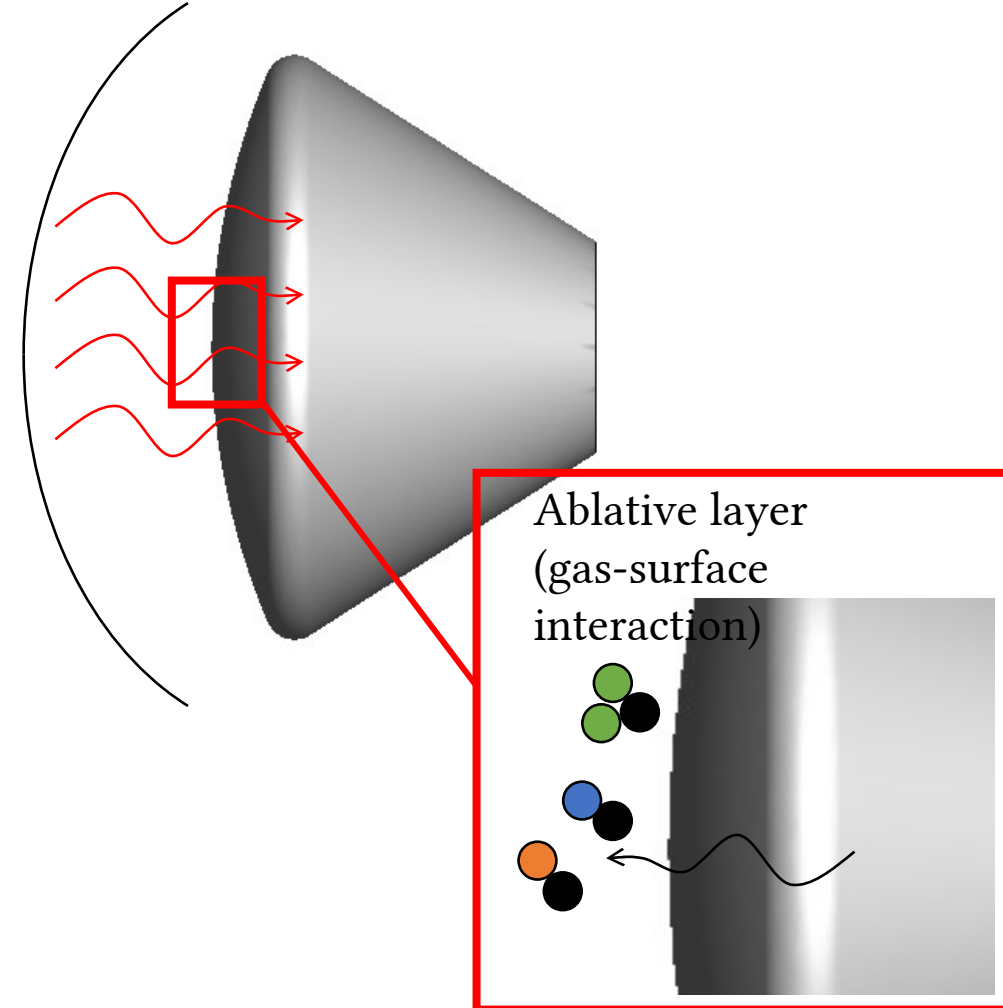
Introduction

- A small fraction is transferred to the surface.

Still

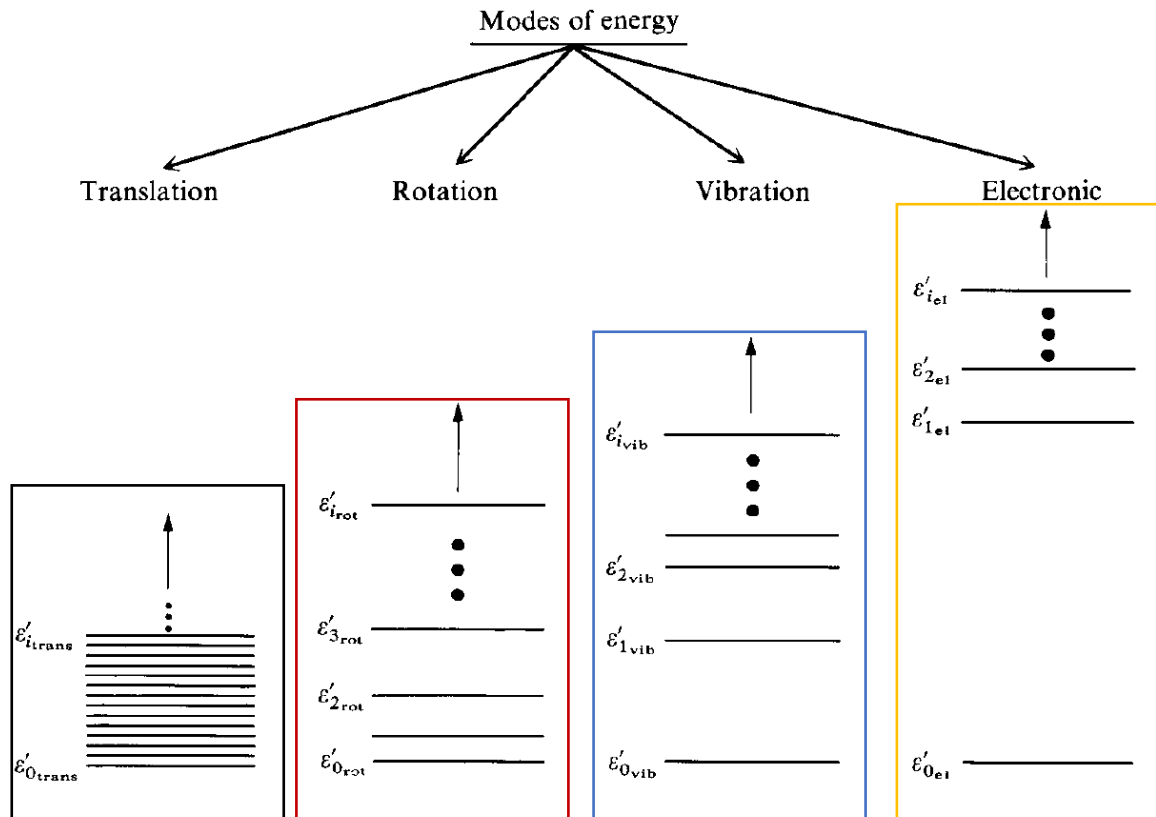
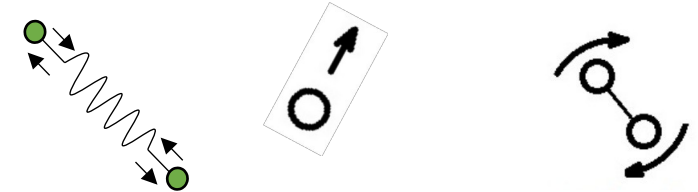
daunting

- Shuttle: 75 W/cm^2
- Apollo: 450 W/cm^2
- Toaster: 0.2 W/cm^2

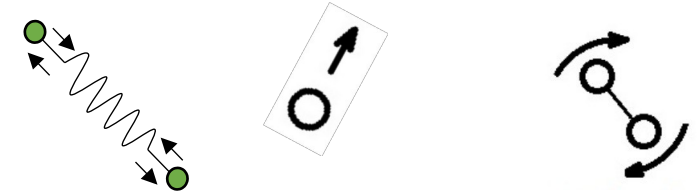


Element of kinetic theory

- Macroscopic world would not exist without microscopic analysis.
- Internal structure is relevant for the definition of the internal energy.
- Microscopic behavior influences the macroscopic behavior of a fluid.



1. Macroscopic world would not exist without microscopic analysis.
2. Internal structure is relevant for the definition of the internal energy.
3. Microscopic behavior influences the macroscopic behavior of a fluid.

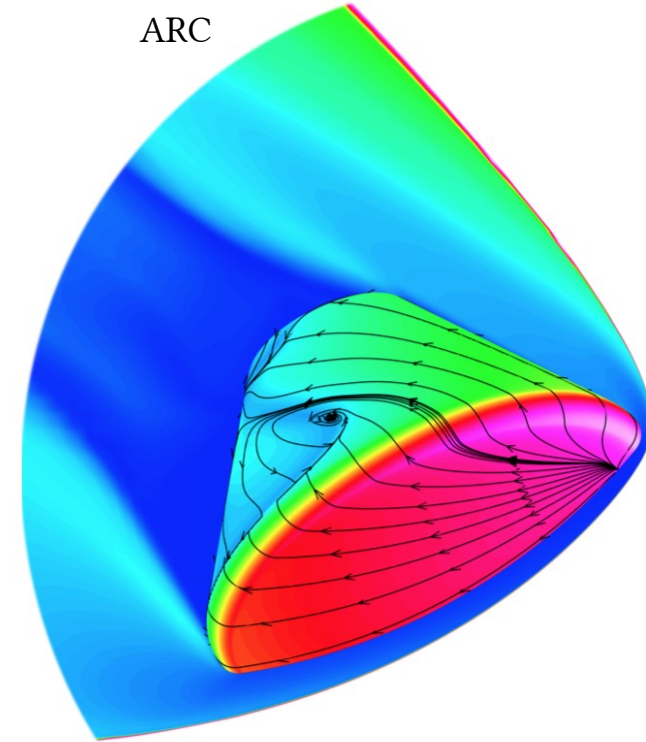


$$\int_{V_0} \frac{\partial}{\partial t} \mathbf{U} dV + \oint_{S_0} (\mathbf{F}_I + \mathbf{F}_V) \cdot \mathbf{n} dS = \int_{V_0} \mathbf{W} dV$$

Perfect gas

Reacting mixture

Credits to NASA
ARC



Hypersonic entry

$$\int_{V_0} \frac{\partial}{\partial t} \mathbf{U} dV + \oint_{S_0} (\mathbf{F}_I + \mathbf{F}_V) \cdot \mathbf{n} dS = \int_{V_0} \mathbf{W} dV$$

Perfect gas

Numerics

Chemistry

Computational
science

Reacting mixture

Applied
mathematics

Transport

$$\int_{V_0} \frac{\partial}{\partial t} \mathbf{U} dV + \oint_{S_0} (\mathbf{F}_I + \mathbf{F}_V) \cdot \mathbf{n} dS = \int_{V_0} \mathbf{W} dV$$

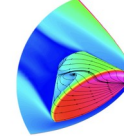
Perfect gas

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix} \quad \mathbf{F}_I^x = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ (\rho E + p)u \end{bmatrix} \quad \mathbf{F}_V^x = \begin{bmatrix} 0 \\ \sigma_{xx} \\ u\sigma_{xx} - q_x \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Reacting mixture

$$\mathbf{U} = \begin{bmatrix} \rho_1 \\ \vdots \\ \rho_{N_s} \\ \rho u \\ \rho E \end{bmatrix} \quad \mathbf{F}_I^x = \begin{bmatrix} \rho_1 u \\ \vdots \\ \rho_{N_s} u \\ \rho u^2 + p \\ (\rho E + p)u \end{bmatrix} \quad \mathbf{F}_V^x = \begin{bmatrix} -\rho_1 u^D \\ \vdots \\ -\rho_{N_s} u^D \\ \sigma_{xx} \\ u\sigma_{xx} - q_x \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} \dot{\omega}_1 \\ \vdots \\ \dot{\omega}_{N_s} \\ 0 \\ 0 \end{bmatrix}$$

Credits to NASA ARC

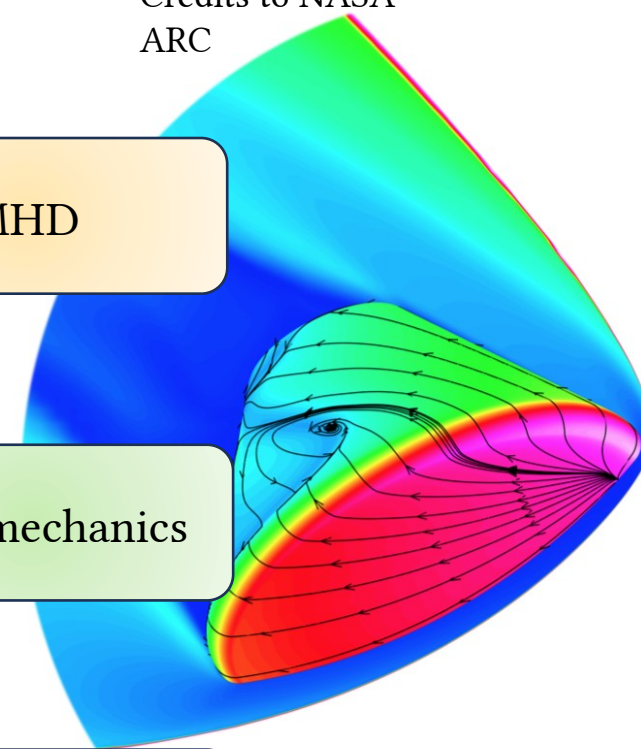


Credits to NASA
ARC

MHD

Solid mechanics

Fluid mechanics

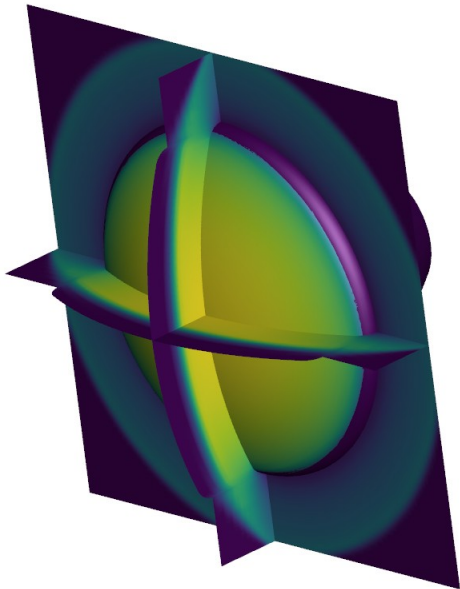


Main goal

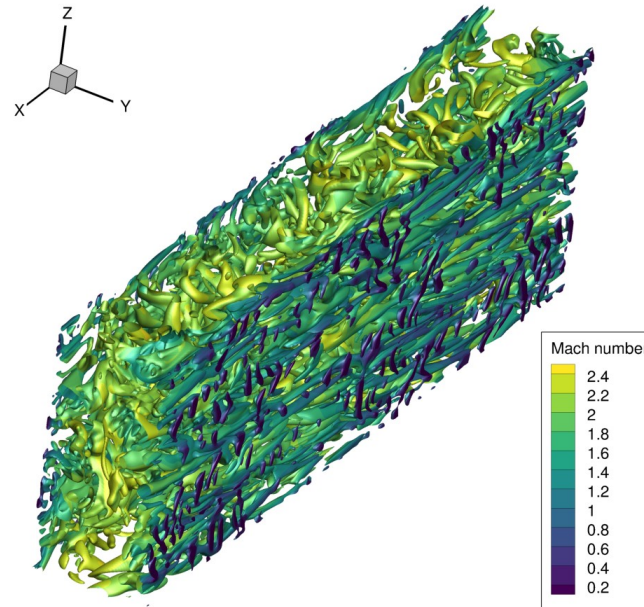
What

1. Reliable prediction of macroscopic quantities of hypersonic flows
2. Development of exascale, efficient CFD tool
3. Simulation of different, complex fluid flows

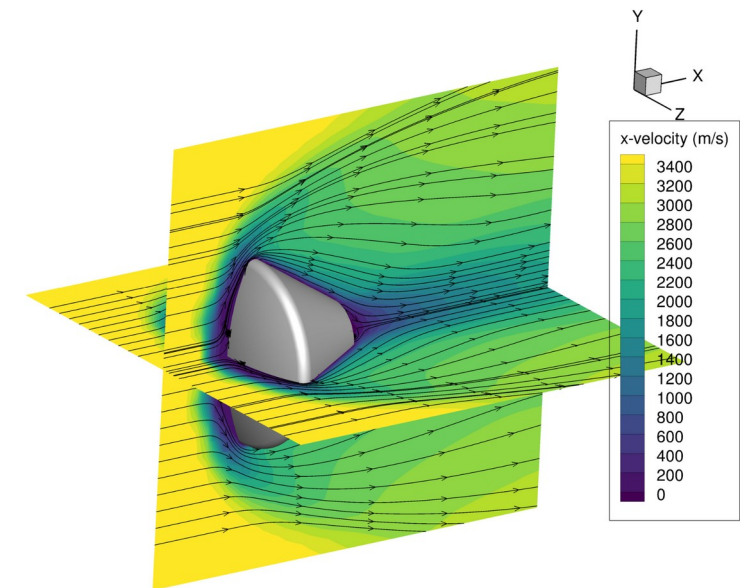
Hypersonic
re-entry



High-order for
turbulent flows



Immersed
boundary



Advection

- Steger-Warming splitting
- Second order MUSCL
- Fifth order WENO
- Sixth order centered Skew-Symmetric scheme

Diffusion

- Second/fourth order conservative flux formulation
- Sixth order Laplacian form

Transport

- Gupta fitting for collision integrals
- Mixing rules for mixture properties
- Fick law for mass diffusion

Time integration

Thermochemistry

- Third order explicit RK scheme (fluid dynamics)
- Park/Splaine/Glass/Seidel scheme (kinetics)
- Rigid-Rotor-Harmonic Oscillator (thermodynamics)



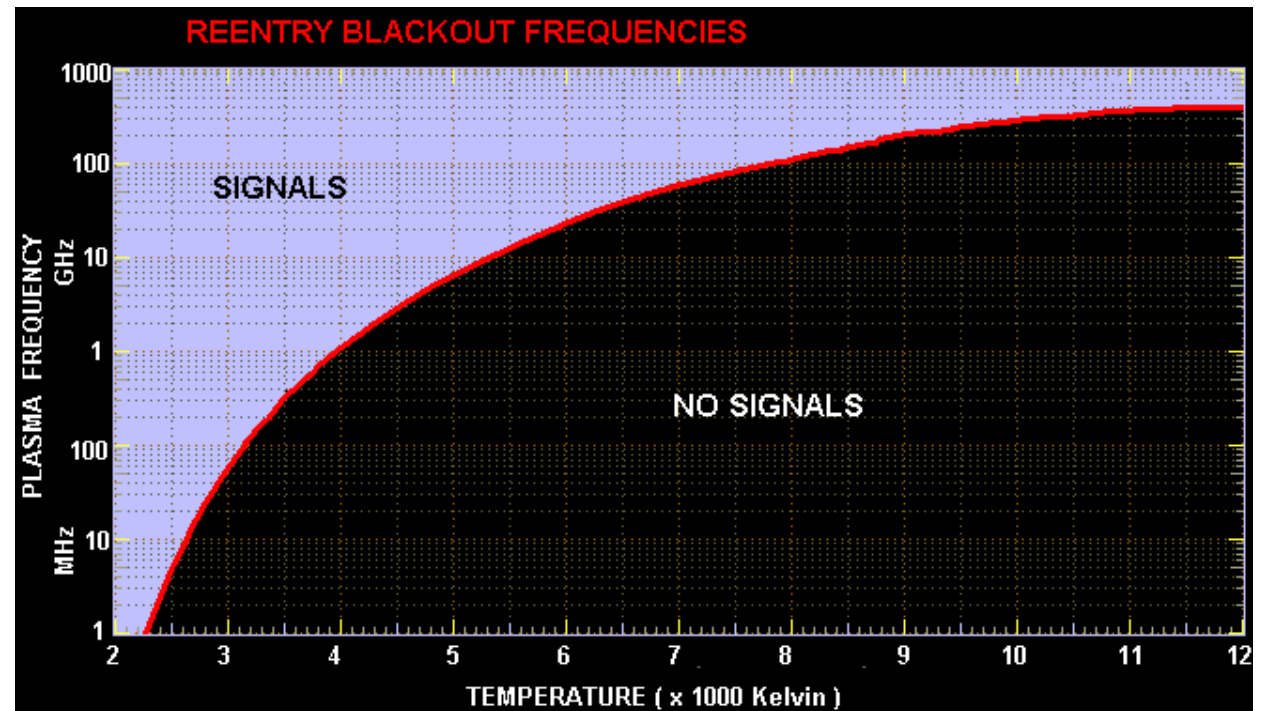
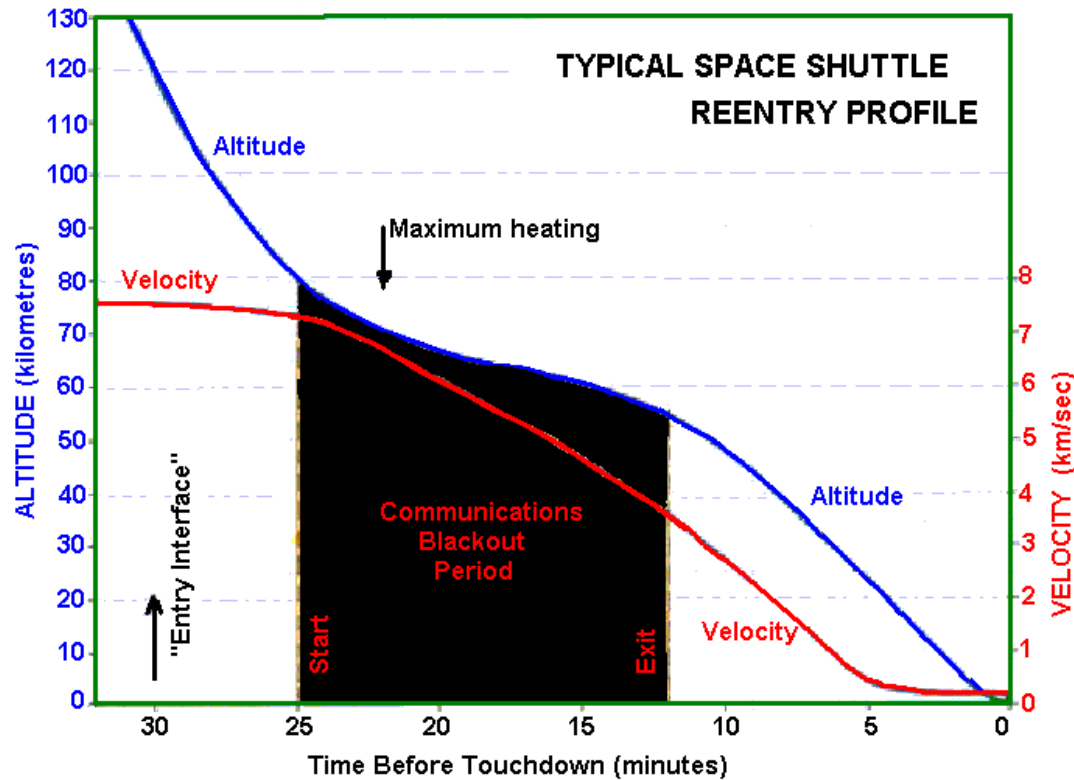
Hardware

- MPI-CUDA (NVIDIA) environment
- High strong scalability
- High weak scalability



Effects of ionization

- Electron density is responsible for **communication blackout**.
- Saha equation is used to estimate ionization surrounding the capsule, basing on number densities on neutral species.



Credits to Australian Space Academy

FIRE II: Flight Investigation of Reentry Environment

- Generation of **aerothermodynamic database** is essential for experimental design of TPSs.
- Code-to-code verifications are useful to assess **different models**.
- High temperature is expected and **radiative heating** is relevant.

Time, sec	Surface Catalycity	Chemistry	DPLR Baseline, W/cm ²	LAURA Baseline, W/cm ²	US3D Baseline, W/cm ²	Largest Percent Difference
1636	Non-catalytic	11-species	183	191	195	7%
1636	Supercatalytic	11-species	301	307	309	3%
1636	Non-catalytic	5-species	226	204		11%
1636	Supercatalytic	5-species	365	340		7%
1643	Non-catalytic	11-species	769	753	762	2%
1643	Supercatalytic	11-species	805	802	797	1%
1643	Non-catalytic	5-species	823	773	787	6%
1643	Supercatalytic	5-species	904	839	857	8%
1651	Non-catalytic	11-species	478	449	455	6%
1651	Supercatalytic	11-species	511	477	506	7%
1651	Non-catalytic	5-species	477	451	457	6%
1651	Supercatalytic	5-species	511	479	506	7%

Project FIRE II launch on May 22nd, 1965



D. Hash et al., AIAA Aerospace Sciences Meeting and Exhibit, 2007

Practical

assumptions...

$$\frac{\partial \rho e_{ve}}{\partial t} = \dot{\omega}_{CV} + \dot{\omega}_{CE} + \dot{\omega}_{VT} + \dot{\omega}_{VV} - \dot{\omega}_I + \dot{\omega}_{eT}$$

- Vibration-Vibration energy exchange is negligible
- Chemistry-Vibration (electronic) coupling is based on production and dissipation of vibrational energy through chemical reactions

$$\dot{\omega}_{CV} = \sum_m e_{vib,m} \dot{\omega}_m \quad \dot{\omega}_{CE} = \sum_s e_{el,s} \dot{\omega}_s$$

- Vibration-Translation transfer represents a collisional term (Landau-Teller)

$$\dot{\omega}_{VT} = \sum_m \rho_m \frac{e_{vib}(T) - e_{vib}(T_{vib})}{\tau_{VT,m}}$$

- The loss in terms of translational energy of electrons is expressed through ionization processes

$$\dot{\omega}_I = \sum_{r \in \mathcal{R}_I} \Delta h_r R R_r$$

- The loss in terms of translational energy of electrons is expressed through ionization processes

$$\dot{\omega}_{eT} = \rho_e \frac{e_e(T) - e_e(T_e)}{\tau_{eT}} = \frac{3}{2} \rho_e R_e \frac{T - T_e}{\tau_{eT}}$$

Practical
assumptions...

$$\frac{\partial \rho e_{ve}}{\partial t} = \dot{\omega}_{CV} + \dot{\omega}_{CE} + \dot{\omega}_{VT} + \dot{\omega}_{VV} - \dot{\omega}_I + \dot{\omega}_{eT}$$

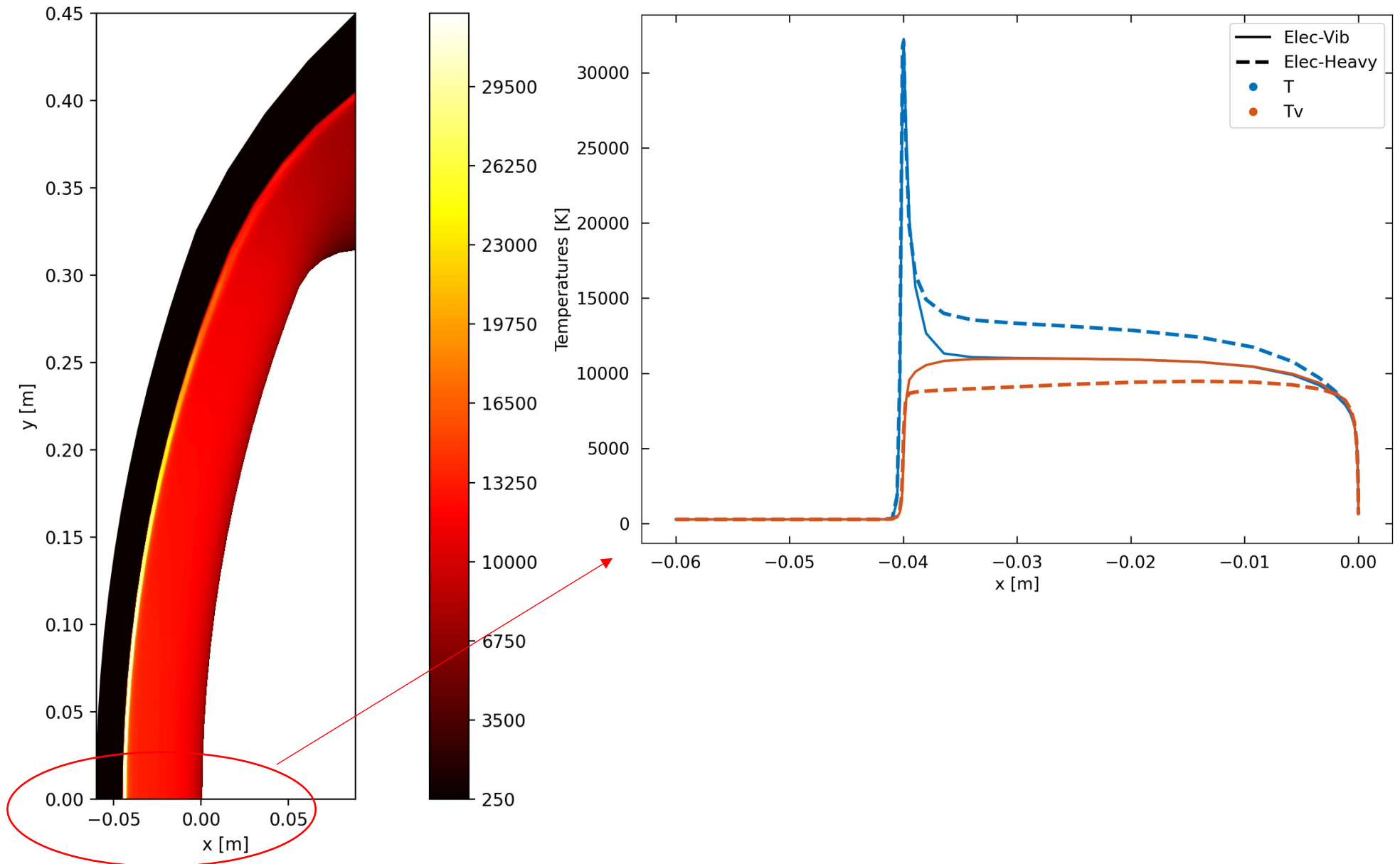
- Common approaches available in the literature assume electrons at equilibrium with the vibronic mode (Elec-Vib).

$$\frac{\partial \rho e_{ve}}{\partial t} = \dot{\omega}_{CV} + \dot{\omega}_{CE} + \dot{\omega}_{VT} - \dot{\omega}_I + \dot{\omega}_{eT}$$

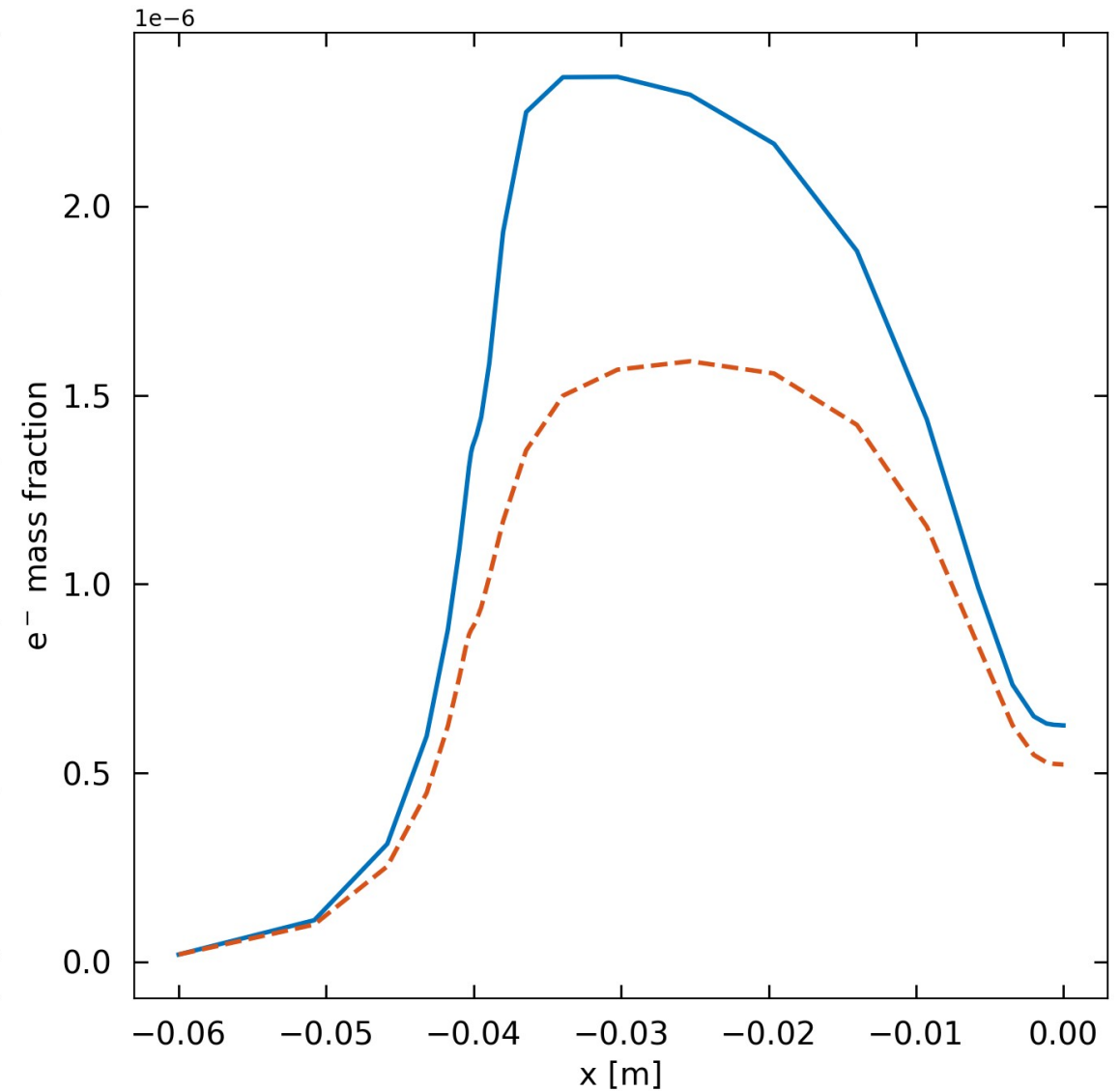
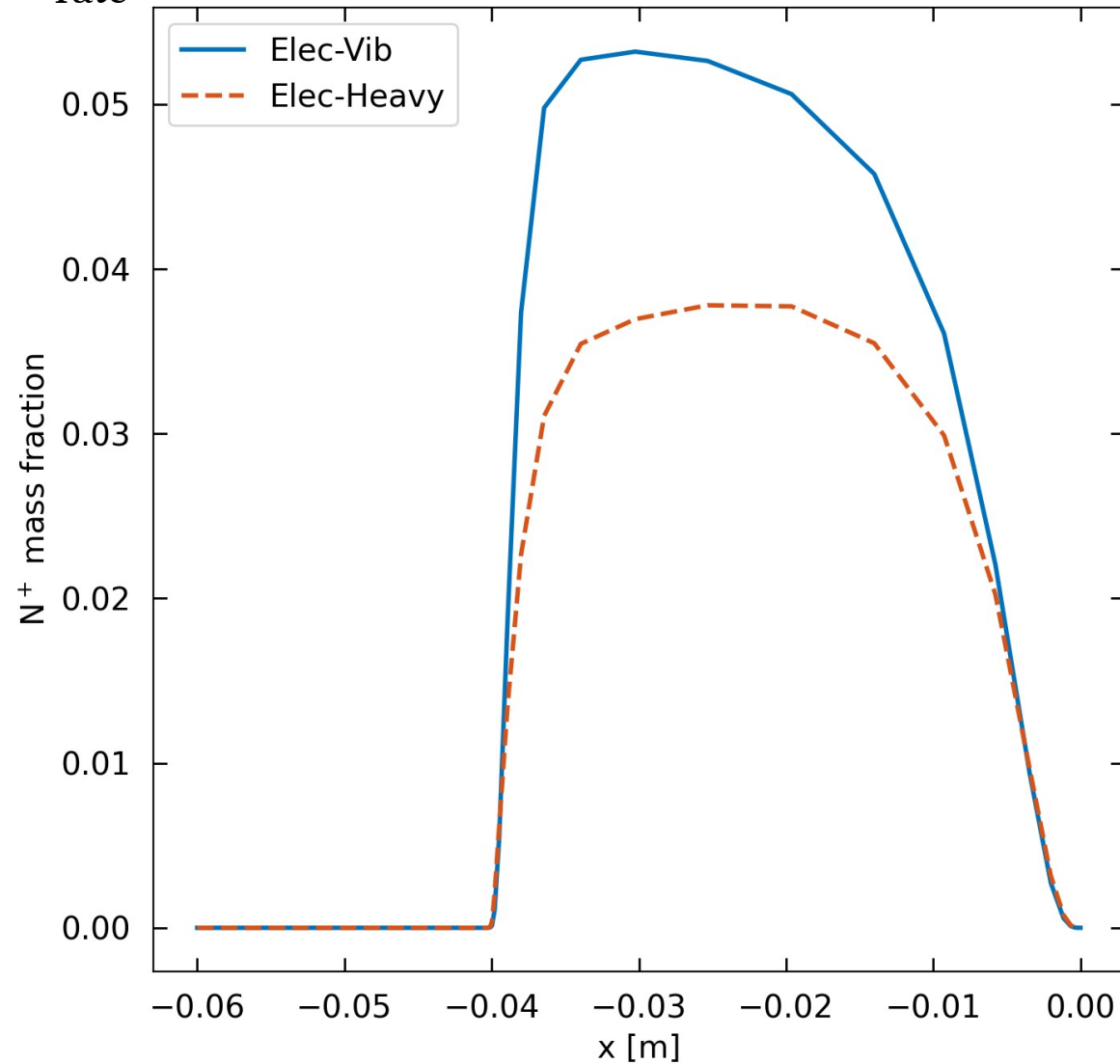
- Electron energy can be defined at the gas temperature (electrons at equilibrium with the heavy species, Elec-Heavy).

$$\frac{\partial \rho e_{ve}}{\partial t} = \dot{\omega}_{CV} + \dot{\omega}_{CE} + \dot{\omega}_{VT} - \dot{\omega}_I$$

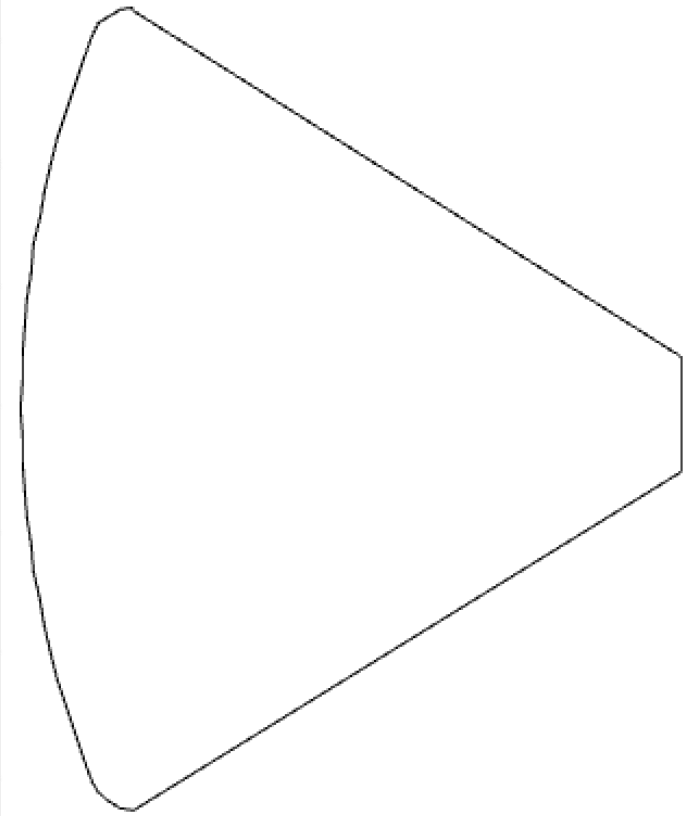
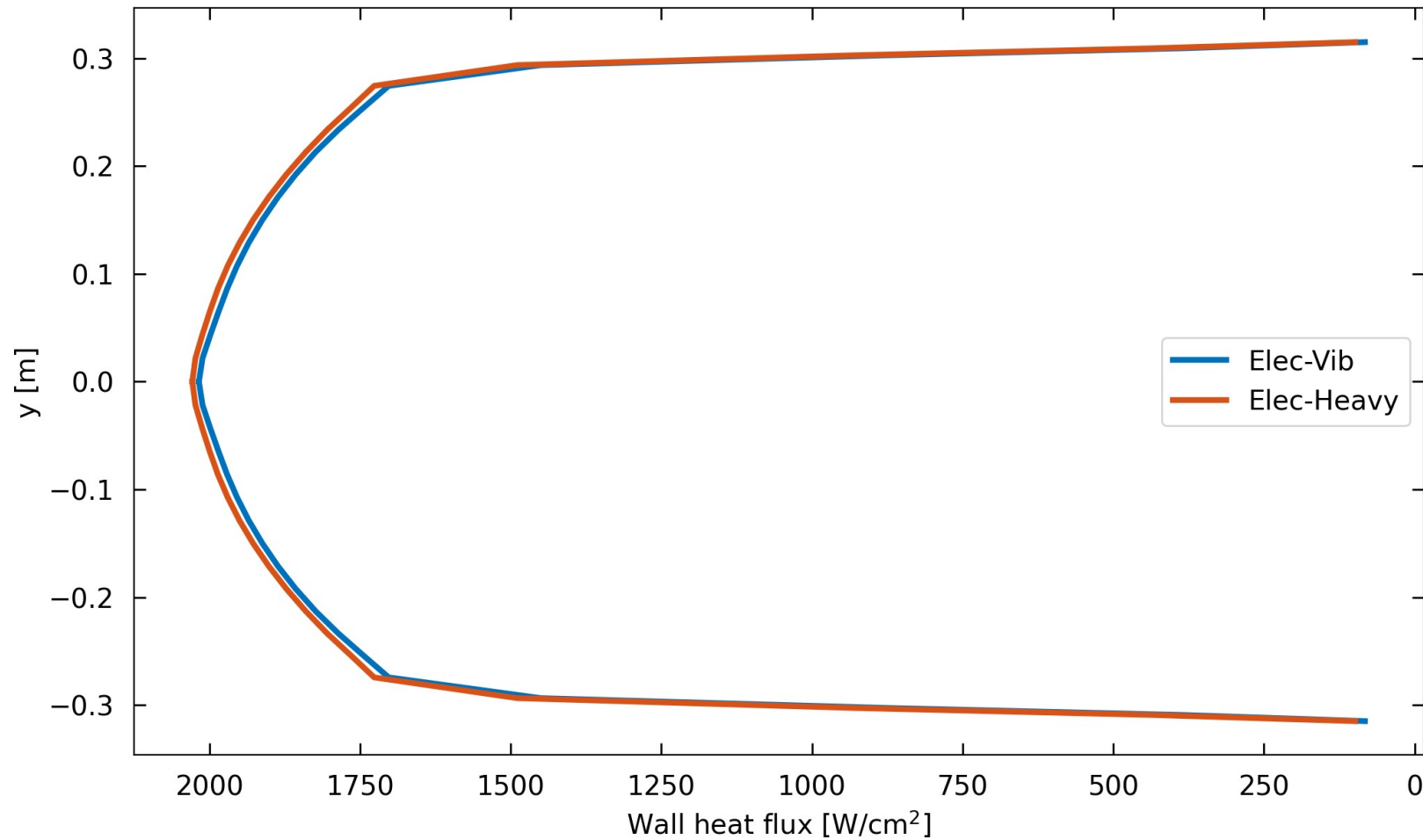
FIRE II: Flight Investigation of Reentry Environment



Different ionization
rate



Negligible differences in the wall heat flux





Modeling is not the only issue



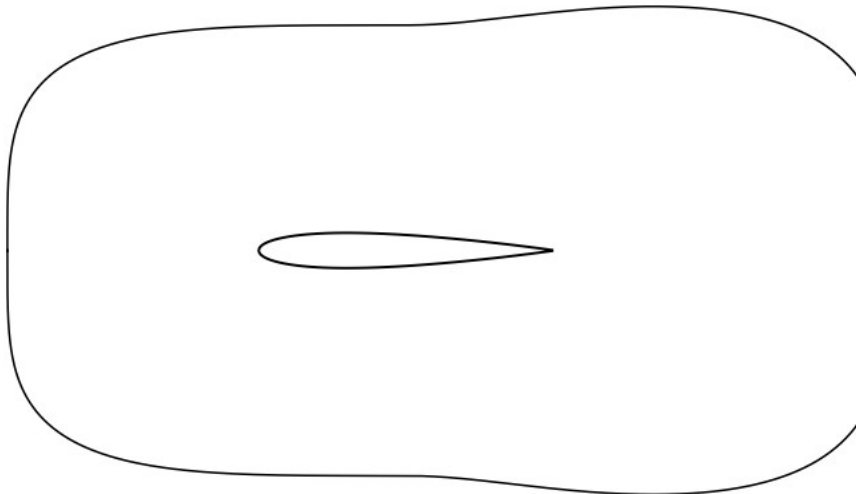
Computational resources



Computational domain



Numerical solver



Johnson Space Center Arc Jet Testing 4" PICA model in CO_2



Remeshing procedure

Complex/moving geometries



Cartesian mesh are much **easier to**



generate.

Fluid and solid domain connected by **interface**



cells.

Suitable for moving (rigid/deformable)



geometries.

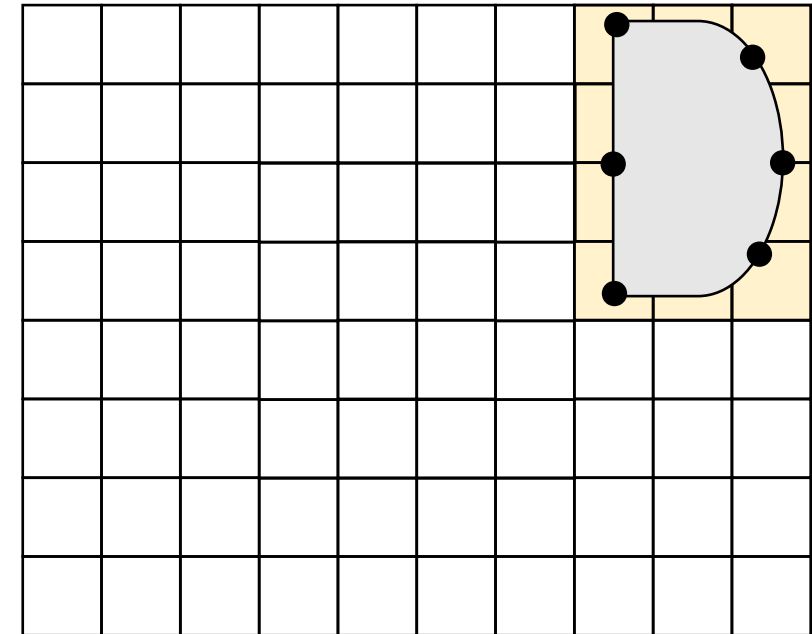
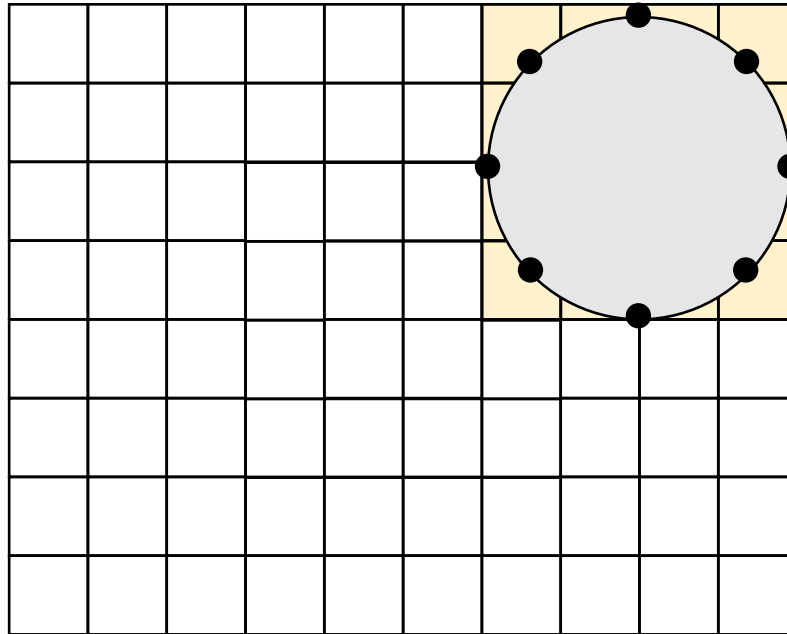
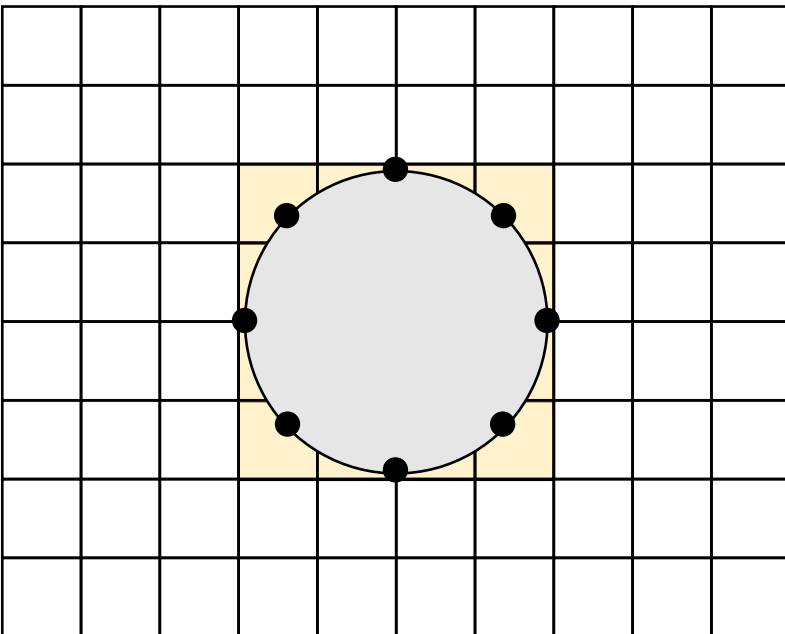
Solid/fluid **forcing strategy.**



Cell aspect ratio.



Wall region accuracy.



Linear interpolation



Easy to implement.
Dirichlet and Neumann.



Linear profiles.
Number of points.
Locally non-conservative.

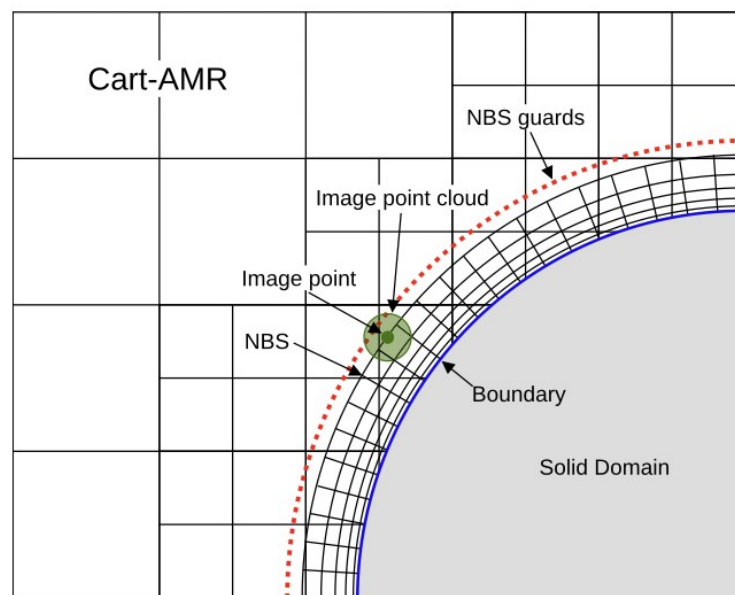
Near-body approach



Body-fitted accuracy.



Meshing effort.



McQuaid and Brehm, Computers and Fluids 269 (2024)

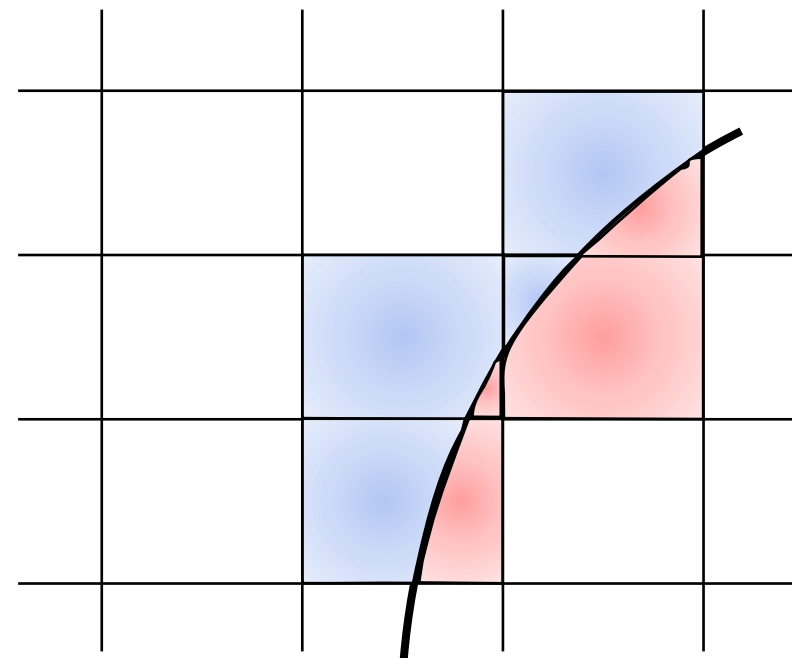
Cut-cell approach



Accurate cartesian.



Complex data structure.



Linear interpolation



Easy to implement.
Dirichlet and
Neumann.



Linear profiles.
Number of points.
Locally non-
conservative.

Boundary
condition
s



Sharp temperature gradients



Mass
conservation



Need of a consistent flux formulation (sharp interface, locally conservative)



Sharp temperature gradients



Mass
conservation



Need of a consistent flux formulation (sharp interface, locally conservative)

Efficiency of immersed boundary methods

- Low enthalpy regime
- Perfect gas
- Low temperature

- Relevant curvature
- High enthalpy regime
- Non-equilibrium
- High temperature

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- Accurate heat flux predictions are **mandatory**
 - a) Immersed boundary efficiency
 - b) Gas-surface interaction modeling
- Multiblock can **extend the capabilities** of the numerical solver (more complex bodies, wake...)
- **Laminar-to-turbulent** transition is critical and can affect the macroscopic features of hypersonic flows
 - a) **Wall-Resolved** Large Eddy Simulations (WRLES)
 - b) **Wall-Modeled** Large Eddy Simulations (WMLES)



Sharp temperature gradients



Mass
conservation



Need of a consistent flux formulation (sharp interface, locally conservative)

Boundary
condition
s

Appendix: IB temperature dependence

- Relevant curvature
- High enthalpy regime
- Non-equilibrium
- High temperature

Isothermal
wall

Perfect gas

Adiabatic
wall

- Relevant curvature
- High enthalpy regime
- Non-equilibrium
- High temperature

Isothermal
wall

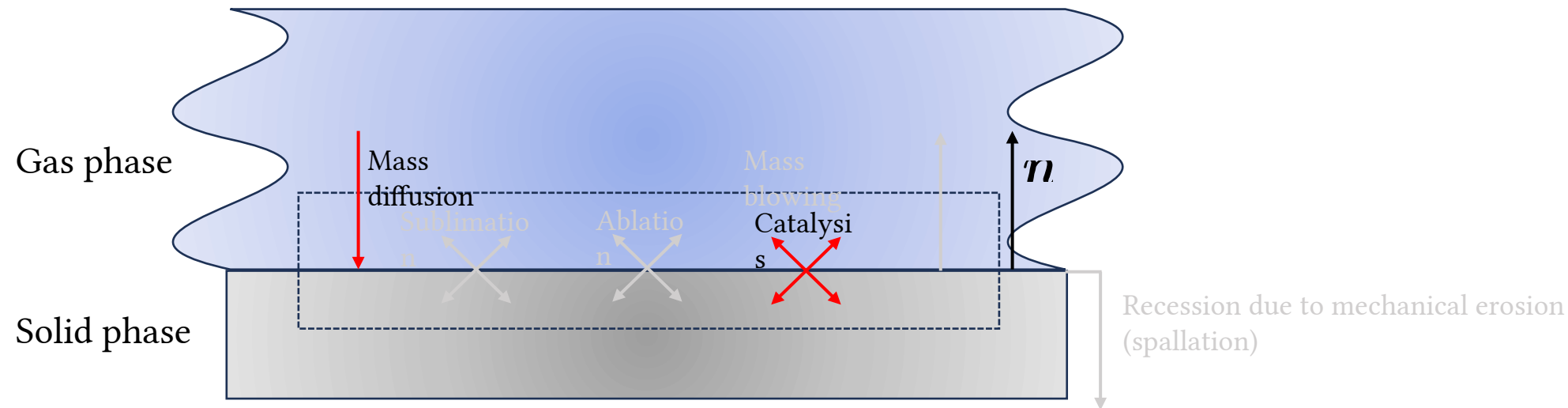
Adiabatic
wall

Appendix: modeling of gas-surface interactions

- **Catalytic materials** promote strong atomic recombination.
- Catalysis is an homogeneous gas-gas mechanism.

- Summing over all the species:

Catalysis does not adduce mass in the mixture



Appendix: modeling of gas-surface interactions

- **Ablative materials** make carbon atoms reach the surface of the TPS.
- Once near the surface, they can sublime or chemically react with the species in the boundary layer ($b = \text{"body"}$).
- As effect of these reactions, the species in the boundary layer are advected away from the surface (blowing effect) with a finite velocity.

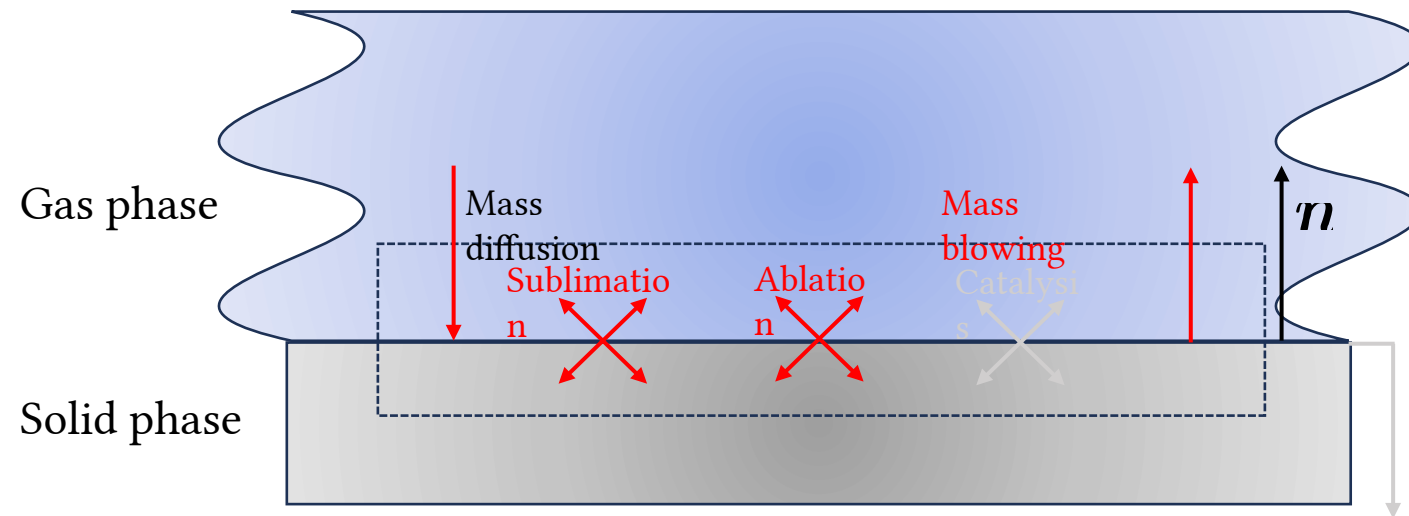
- Summing over all the species:

where

Ablation adduces mass due to the massblowing rate.

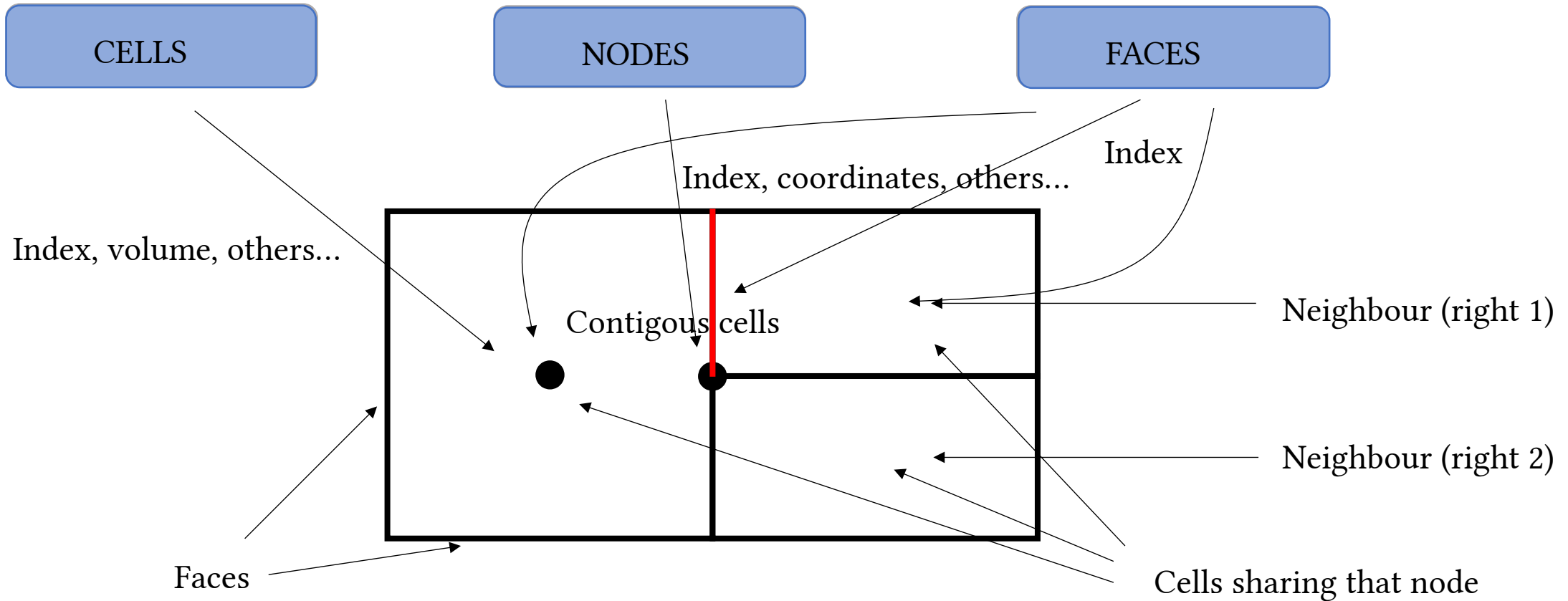
In the case of recession

Recession due to mechanical erosion (spallation)



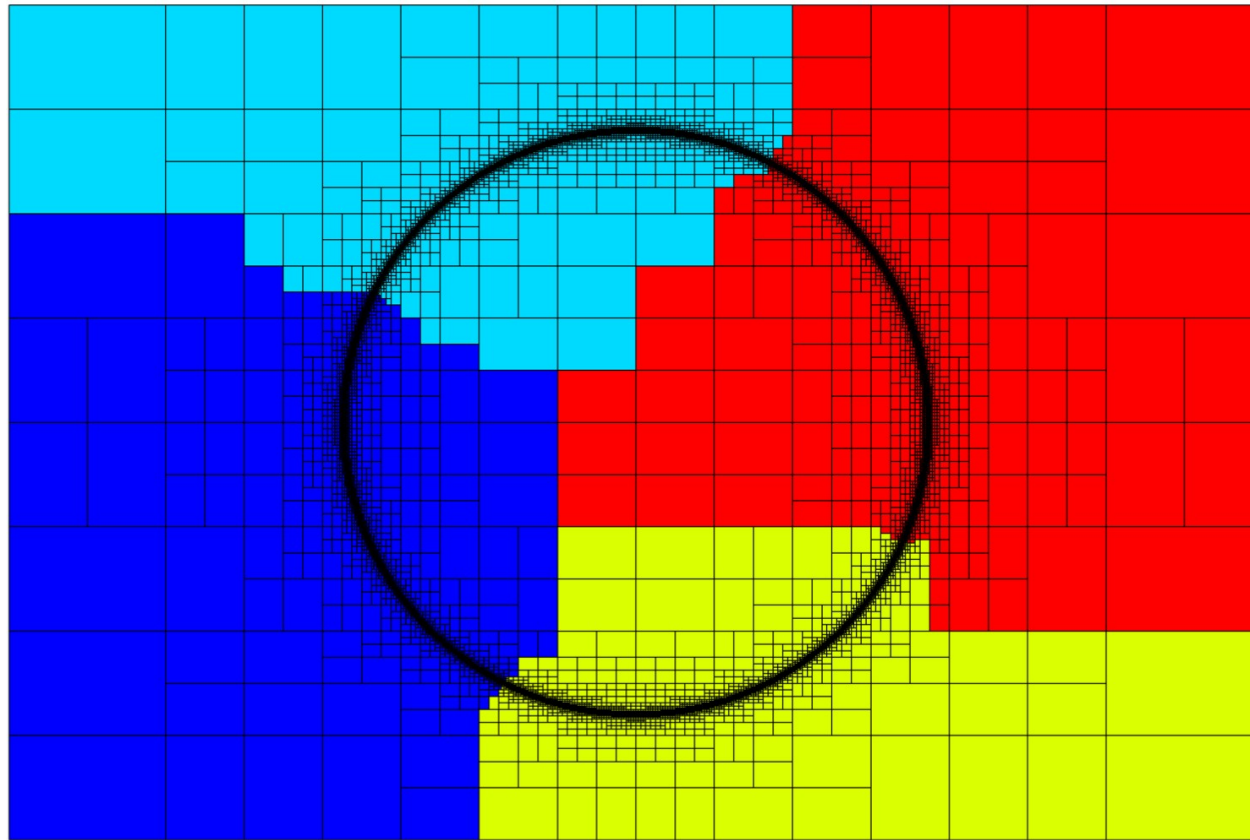
Appendix: IB data structure and connectivity

TOMMIE (*Tired-of-Meshing*) is used for the mesh generation.



Each partition of a regularized domain boundary is a foldable strategy for an equal cells balance has to be thought.

METIS is a tool for graph partitioning.



Since the grid is unstructured, the partitioning can not create a regular topology.

domain ghost cells (in this

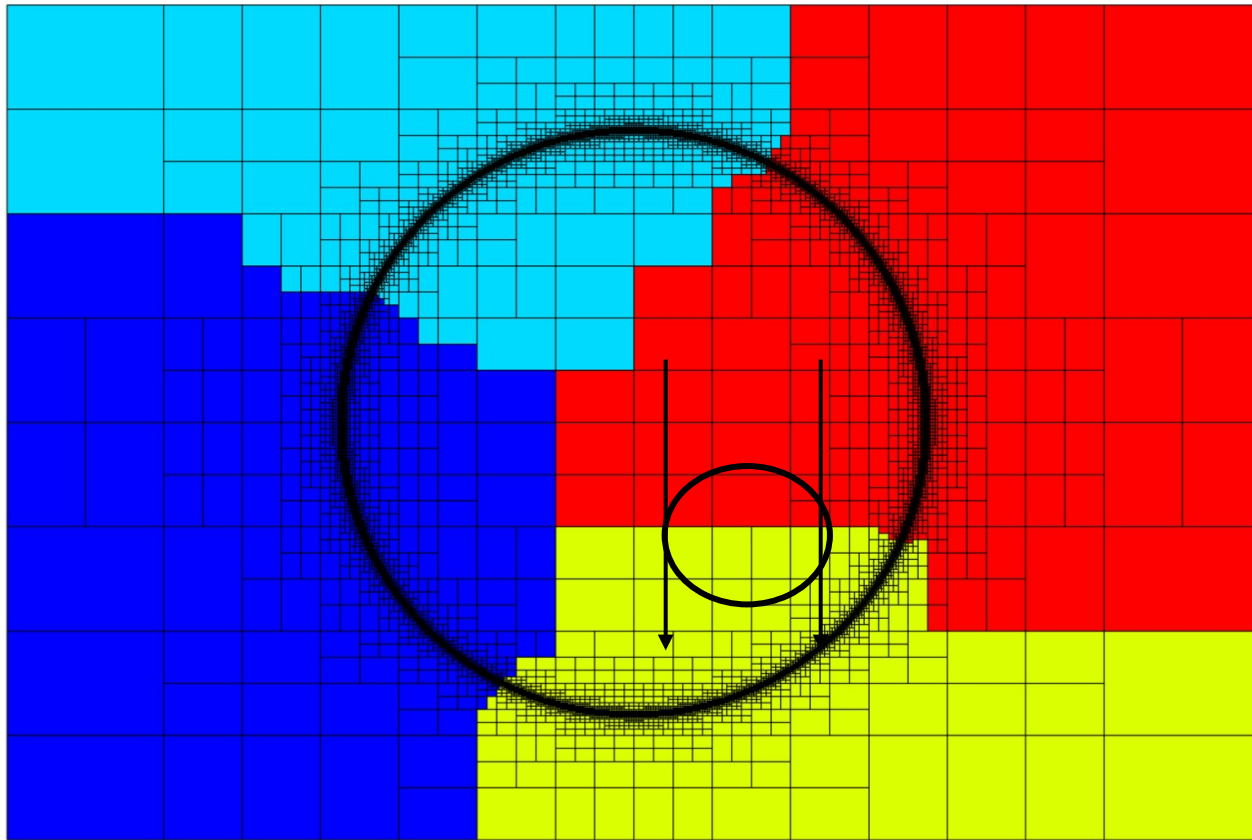
example, one ghost cell)

Need of readapting the connectivities

information for local computations (processors).

What about ghost cells? The number of ghost cells basically depends on the implemented solver.

The ghost cells are needed also at the boundaries between processors.



In this case the red cell needs two ghost cells at the bottom.

Nevertheless, one can think about more complicated configurations. So an efficient searching algorithm has to be implemented.

